

inadvertently contain outdated or opinionated content too, people have started becoming aware of the problems in AI—from safety, security, misinformation, and privacy issues to bias and discrimination. No wonder, the year seems to be ending on a more cautious note, with nations giving a serious thought to the risks and required regulations, not as isolated efforts but collaboratively. This is because, like the internet, AI is a technology without boundaries and a combined effort is the only possible way to control the explosion.

Tech, thought and political leaders from across the world met at the first global AI Safety Summit, hosted by the UK government, in November. The agenda was to understand the risks involved in frontier AI, to build efficient guardrails, to mitigate the risks, and use the technology constructively. The summit was well-attended by political leaders from more than 25 countries, celebrated computer scientists like Yoshua Bengio, and technopreneurs like Sam Altman and Elon Musk.

Frontier AI is a trending term, that refers to highly capable general-purpose AI models, which match or exceed the capabilities of today's most advanced models. The urgency to deal with the risks in AI stems not from the current scenario alone, but from the realisation that the next generation of AI systems could be exponentially more powerful. If the problems are not clipped at the bud, they are likely to blow up in our faces. So, the summit was an attempt to expedite work on understanding and managing the risks in frontier AI, which include both misuse risks and loss of control risks.

In the run-up to the event, UK's Prime Minister Rishi Sunak highlighted that while AI can solve myriad problems ranging from health and drug discovery to energy management and food production, it also comes with real risks that need to be dealt with immediately. Based

on reports by tech experts and the intelligence community, he pointed out several misuses of AI, ranging from terrorist activities, cyber-attacks, misinformation, and fraud, to the extremely unlikely, but not impossible risk of 'super intelligence,' wherein humans lose control of AI.

The first of what promises to be a series of summits, was characterised mainly by high-level discussions, and countries committing themselves to the task. Representatives from various countries, including the US, UK, Japan, France, Germany, China, India, and the European Union signed the Bletchley Declaration. They acknowledged that AI was rife with short-term and longer-term risks, ranging from cybersecurity and misinformation, to bias and privacy; and agreed that understanding and mitigating these risks requires international collaboration and cooperation at various levels.

The declaration also highlighted the responsibilities of developers. It read—“We affirm that, whilst safety must be considered across the AI lifecycle, actors developing frontier AI capabilities, in particular those AI systems which are unusually powerful and potentially harmful, have a particularly strong responsibility for ensuring the safety of these AI systems, including through systems for safety testing, through evaluations, and by other appropriate measures.” Sunak is also said to have made a high-level announcement about makers of AI tools agreeing to give early access to government agencies to help them assess and ensure that they are safe for public use. At the time of this story being drafted, we still have no information of what level of access is being referred to here—whether it would be just a trial-run, or code-level access.

Regulations, research, and more

The UK government also launched the AI Safety Institute, to build the

intellectual and computing capacity required to examine, evaluate, and test new types of AI, and share the findings with other countries and key companies to ensure the safety of AI systems. This institute will permanentise and build on the work of the Frontier AI Taskforce, which was set up by the UK government earlier this year. Researchers at the institute will have priority access to cutting edge supercomputing infrastructure, such as the AI Research Resource, an expanding £300 million network comprising some of Europe's largest supercomputers; as well as Bristol's Isambard-AI and Cambridge-based Dawn, powerful supercomputers that the UK government has invested in.

On October 30th, US President Joe Biden signed an executive order that requires AI companies to share safety data, training information, and reports with the US government prior to publicly releasing large AI models or updated versions of such models. The order specifically alludes to models that contain tens of billions of parameters, trained on far-ranging data, which could pose a risk to national security, the economy, public health, or safety. The executive order emphasises eight policy goals on AI—safety and security; privacy protection; equity and civil rights; consumer protection; workforce protection and support; innovation and constructive competition; American leadership in AI; and responsible and effective use of AI by the Federal Government. The report also suggests that the US should attempt to identify, recruit, and retain AI talent, from amongst immigrants and non-immigrants, to build the required expertise and leadership. This has gained some attention in the social media, as it bodes well for Indian tech professionals and STEM students in the US.

The standards, processes, and tests required to enforce this policy will be developed by government agencies